Selecting and Planning Good Research Projects

Chapter objectives

After reading this chapter you will be able to:

- **Generate new ideas for research.**
- **Identify a good research topic using selected criteria.**
- **Write a proposal for the research project.**
- **Plan the research project and data collection.**
- **Gain the access you need to data.**
- **Conduct research in an ethical manner.**

Generally, research projects can be designed as part of an academic programme of study, or as a result of a business or organizational need. While the former will probably require a stronger element of theoretical under-pinning, both will need a sharp, practical focus or application. The outputs from research projects not only have potential benefits for organizations and their manage-ment, they can also be a vital element in personal learning and development. Clearly, the best approach is to select a research topic that interests you, and one that is likely to maintain your interest. The research process can be a long and arduous one, so you need to be committed to your subject.

Winkler and McCuen (1985) suggest that you also need to select a subject area that has sufficient scope to generate several research projects. So, for example, investigating how a particular commodity is produced at source and shipped to a retail outlet may not prove particularly illuminating. On the other hand, a study of supply logistics in general may offer greater scope for the research process, the range of literature to be con-sulted and the value of the research outcomes.

If you find you have difficulty finding a research subject, then talk to colleagues at work to see what sort of issues concern them. Discuss the matter with your academic supervisor or line manager. Other useful sources are professional

journals and magazines that often contain articles on issues that are currently engaging the minds of business, commerce, public sector and voluntary organizations. You might also browse through the business or management sections of your local bookshop to see what kinds of titles are being published. Some more practical suggestions for generating ideas are given in this chapter. Some advice is also given on how to select a research topic and on how to write a successful proposal for your research. It also suggests how you can plan a schedule for conducting your research and how you should carry out the project, not only efficiently, but ethically.

SELECTING A RESEARCH TOPIC

You may already have a research topic in mind and hence want to use this section as a means of checking its validity. Alternatively, you may have been commissioned by your organization to undertake a specific piece of research. In the latter case, do not feel that you should be a passive recipient of such projects. Make use of the criteria in this section to evaluate and renegotiate the focus of your project if you feel that this is necessary.

When to select a research topic

Obviously, this is going to be a matter of individual choice. Some researchers have a very clear idea and focus at an early stage. Indeed, they may have embarked on a programme of study precisely because they want to tackle a specific end project. For others, and probably the majority, a research topic emerges only towards the end of the study programme, or as a result of an emerging problem in the workplace. For some, the problem may be making a choice between a number of potential topics; for others, there may be only one focus.

Sources of research topics

There are, essentially, two ways of identifying a research topic. One is through the literature – books and academic and professional journals – which may raise interesting themes and topics that can be related to your own organization. The other route is directly from the workplace or community setting. Line managers, supervisors or project teams may all require assistance, and this can often be a fruitful source of research focus. In effect, the researcher then acts as a kind of internal consultant to a project team.

What is a good research topic?

Whatever topic you choose, it is likely that you will begin to develop or enhance a range of personal skills. A good topic, then, is one that gives you free rein to

maximize this self-development. Jankowicz (1991) argues that such personal development might include:

- Improving personal time management.
- Gaining access to respondents.
- Interviewing respondents.
- Speaking to an audience.
- Persuading people to cooperate.
- Dealing with uncertainty about data.

But it must also be a subject that interests you. Since research may involve many hours of planning, execution, data analysis and report writing, you will quickly tire of any topic that you thought was only moderately interesting at the outset. It is also a good idea to choose a subject that allows you to demonstrate your skills and abilities. Hence, if, say, you are undertaking a project at the end of an academic programme, you will need to select a subject that gives you scope for showing the integration of various fields of knowledge and analysis. Within the workplace, being able to demonstrate the skills of planning, data analysis and report writing can enhance your prestige and even promotional opportunities.

Academic requirements

You must ensure that the research subject is capable of meeting academic requirements if you are undertaking a programme of study. As Raimond (1993) suggests, be sure that your topic is capable of being linked to the appropriate academic theory. Management theory, for example, tends to evolve and change quite quickly. Textbooks, however, can take years to write and are often out of date by the time they are published. One solution is to look at the academic journals (many of which are now online), which tend to be more topical.

Access

You will need access to relevant information, material and data. If you select an issue where these are lacking, you have little chance of completing the project. Remember that some issues in organizations, communities or networks are sensitive or confidential, for example, some financial data, redundancy plans, attitudes within a community, etc. Indeed, Flick (1998) warns that a research project is an intrusion into the life of an institution and is inherently unsettling for it. Apart from written or Web-based information, one of the essential elements of research is access to people. Who will you need to meet (perhaps to interview) and how busy are they? A classic contradiction is that the more important your project (to the organization), the more likely it is that the people you need to see are senior in the organization, and too busy to spare the time to be interviewed. The challenge is to gain access to these people despite this. In community settings, the researcher may be seen as an intruder or outsider.

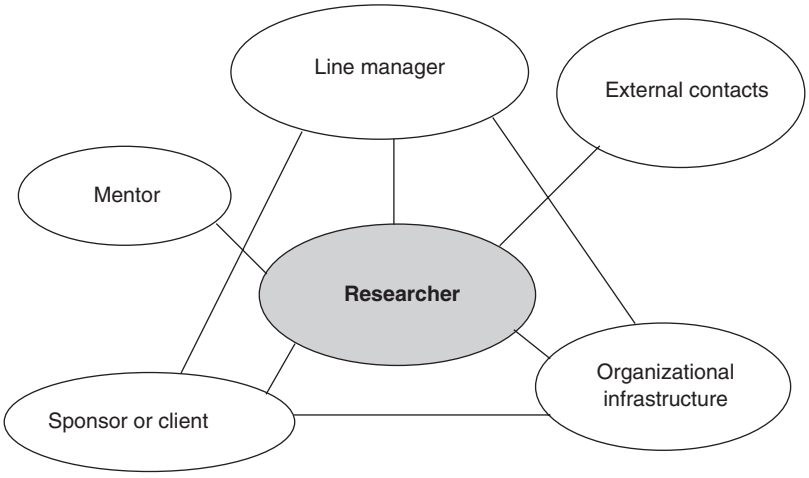


FIGURE 3.1 ORGANIZATIONAL SPONSORSHIP AND SUPPORT NETWORKS

Sponsorship and networking

It helps if you have a sponsor or client who can give you either financial backing, or at least moral or practical support. The latter might involve 'opening doors' in the organization and facilitating your access to people with information. Figure 3.1 shows the kind of networks that may exist, or which you may request are established, to provide you with assistance. Note that not all elements of this network are necessarily connected. They all perform different roles, so you need to understand or negotiate what each can offer you.

Activity 3.1

Make a list of the support networks available to you. Are they readily accessible? Are they sufficient?

Time available

Be sure that the research can be completed within the time available. There is always a tendency to underestimate the time needed for a project. Further difficulties may arise if the topic chosen is dependent upon the implementation of another project within the organization. If this project becomes delayed (which is often the case), or abandoned, then your research project may quickly reach an impasse. The best approach is to draw up a research plan before starting the project, with clear indications of dependencies and potential bottlenecks (see Figure 3.3).

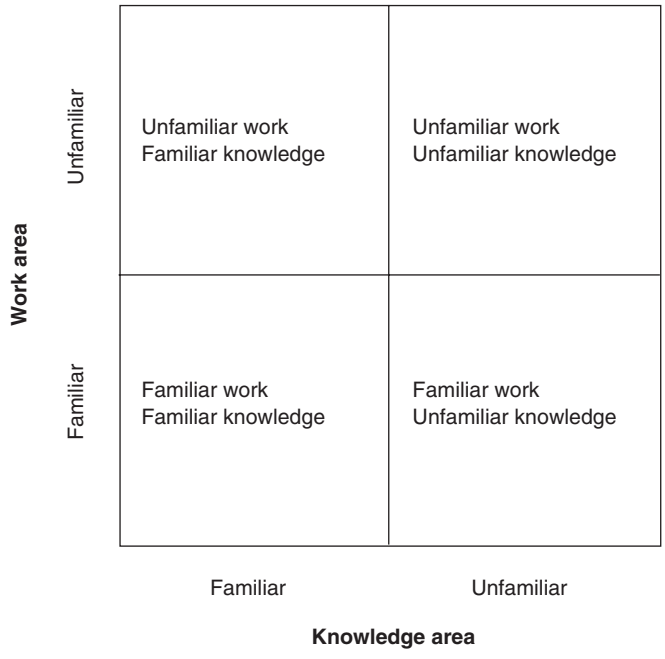


FIGURE 3.2 JOHARI WINDOW SHOWING CHOICES BETWEEN FAMILIAR AND UNFAMILIAR WORK EXPERIENCE AND PERSONAL KNOWLEDGE

Capabilities and experience

This may seem obvious, but selecting an issue that is within your capabilities is essential. Your skills will, hopefully, develop during the course of the research process; but, say, choosing a topic that requires robust statistical skills when you are comfortable with only basic mathematics may be a recipe for disaster. Experience may be a different matter. As the Johari window in Figure 3.2 shows, you can choose projects that are congruent with both your work area and experience (the safe approach), or, moving towards the top-right side of the diagram, beyond both your work and current knowledge set. This poses greater risks, but also enhances opportunities for personal development. Moving the project into an unfamiliar work area may also provide opportunities for networking amongst new groups of people which can be advantageous for both the project and your own professional future (including your future as a researcher).

Activity 3.2

Take the project that you intend to do, or one or more projects that you are considering. Locate the position of the project(s) within the Johari window. How risky is the project, and are the risks worth taking? How 'stretching' is the project and is such development a personal objective?

Value of the project

Projects that have value to the organization (say, in terms of identifying cost savings, new marketing opportunities, IT strategies, etc.) will have a much greater chance of success than those that merely re-plough old ground. Innovative, provoking and original projects have a better chance of sponsorship and support from within the organization (at the appropriate level), of opening new networks for meeting and interviewing people, and of eventual implementation. A high value project is also more likely to motivate both you and your line manager or sponsor. But ambitious projects may be more difficult to manage and complete. Sometimes, a more modest project may be both more feasible and achievable.

Symmetry of potential outcomes

Gill and Johnson (1997) suggest that one way of reducing the risks involved in a project is to achieve symmetry of potential outcomes. This means that, no matter what the results are, they will be useful. For example, a project to examine whether the company's e-commerce website produced any increase in overall sales would be significant whatever the outcome. Conversely, a project that examined the relationship between levels of stress and output levels amongst line workers would be interesting if strong correlations were found, but would be of little value if they were not.

Career goals

You may consider whether the research topic may be of value to you in the future in terms of your personal career development. The research may make you an 'expert' in a particular subject or area, and enhance your value as an internal consultant.

Generating research ideas

If you are devoid of ideas for a topic, how can you create some imaginative ones? Table 3.1 suggests that new ideas can be generated either through rational or creative processes. Let us look at each of these in turn.

Examining your own strengths and weaknesses

You will benefit from choosing a topic that you enjoy and for which you have probably received good marks for previous assignments and other course assessments, or positive feedback from a work-based project. Why not make a list of your strengths and weaknesses. Get a friend or colleague to critique the list (prepare yourself for a shock!), then amend it as necessary.

Looking at past projects

This is often a useful way of generating new ideas. A glance towards the end of some projects may reveal a section entitled 'Suggestions for future research' that

TABLE 3.1 TECHNIQUES FOR GENERATING AND REFINING RESEARCH IDEAS

Rational thinking	Creative thinking
Examining your own strengths and weaknesses	Brainstorming
Looking at past projects	Exploring personal preferences using past projects
Searching the literature	Relevance trees
Discussion	Keeping a notebook
	SWOT analysis

Source: Adapted from Saunders et al., 2000

may be helpful. There may also be a bibliography which could prove a useful starting point for your own research – although take care that the references are not too dated. Also note that some universities and colleges place *all* theses and dissertations in the library. Their presence there, then, is not a necessary guide to their quality.

Searching the literature

The literature includes articles in the academic journals, reports, books and websites (although be wary of the authenticity and quality of the latter). Becoming aware through the literature of the significance of some issues, or new angles on old ones, can be a stimulus to undertake research in how these ideas can affect your own organization. More detail on reviewing the literature is provided later in this chapter.

Exploring personal preferences using past projects

Here, you simply take a look at the subjects you have chosen for previous modules, programmes or work projects, and identify the kinds of topic areas you have selected. This may be a guide to what you are generally interested in.

Discussion

Ideas might be generated by talking to fellow students, work colleagues, line managers, university tutors, practitioners and professional networks (the latter, possibly, through online discussion groups).

Brainstorming

This is a well-known problem-solving technique for generating and refining ideas. Jarvis (1995) suggests that the quantity of ideas produced by the brainstorming group is more important than the quality. All points made by participants are recorded (for example, on a flipchart) over a mutually agreed period of time. No member of the group may criticize the ideas of another, irrespective

of how ridiculous some ideas may sound, since this would inhibit discussion. At the end of the agreed time period, the group discusses and selects from the points raised.

Relevance trees

This is similar to mind mapping, where you start with a broad concept from which you generate more specific topics. From each of these branches, new sub-branches can be generated. Do this quickly with an emphasis on generating ideas rather than evaluating them for quality. Once the process is finished, look over your material and evaluate the results.

Keeping a notebook of ideas

This simply involves noting down any new ideas as and when they occur. It is best to keep the notebook with you at all times. This could be a section in a research log book (see Managing information, p. 46).

SWOT analysis

SWOT stands for Strengths, Weaknesses, Opportunities and Threats. Using this well-known method, you could make a list of ideas under each of these categories. SWOT analysis usually works best, however, when undertaken by a group since good ideas tend to generate others.

Activity 3.3

Begin the process of generating ideas for your project. Note down fresh ideas as they occur. This may be when reading the literature, talking to people at work, or talking to people undertaking a similar programme of study. At some point, you may chose to carry out a SWOT analysis on each subject to see which of them has the most potential.

TOPICS TO AVOID

It is often only possible in retrospect to recognize the topic you should not even have attempted! However, here are a few hints that may help you to avoid the research disaster. The topics to avoid are those that are:

- *Too big*. For example, 'Human resource management – innovative international perspectives'. Some very large projects can be worthy and valuable to an organization, but you need to ask yourself whether you have the time,

experience and resources to complete them. Winkler and McCuen (1985) also warn that the big topic is also the most difficult to write about: it is difficult knowing where to begin, and omissions and oversights are more crudely exposed.

- *Traced to a single source*. This may not be a particular problem in pure business research when a single solution is needed to a problem. However, if the research is linked to an academic programme of study, or important for your own professional development, there will usually be a requirement that issues are explored from a variety of different angles.
- *Too trivial*. This may seem rather subjective, but you should use your common sense to evaluate the kinds of projects that are worth doing and those that are not. As a general rule of thumb try using the 'So what?' test. Ask yourself, after completing the research, whether the results have any meaning or significance (to others not just to yourself). For example, a research project that surveyed how to reduce the use of paper in a marketing department of ten people would yield very little of value. On the other hand, a project that took the issue of recycling (paper, printer cartridges, furniture, computers, etc.) across an organization could have considerable scope and link into the broader environmental debate.
- *Lacking in resource materials*. Look out for warning signs – very few references to the topic in the main textbooks, practitioner journals or other refereed journals or websites. If the project is going to rely on access to in-house knowledge experts, make sure (in advance) that they are both available and willing to cooperate with you.
- *Lacking in sponsorship*. This does not necessarily mean financial sponsorship, but it is often important to obtain the support and commitment of key people in the organization or fieldwork setting where the research is taking place. These are likely to be directors, senior managers, or the leaders of networks or groups.
- *Too technical*. Some projects are more concerned with solving highly technical problems rather than organizational research. Leave these to the technical gurus.
- *Intractable*. You may be offered a problem that nobody else has been able to solve. Be highly suspicious of this kind of gift! Ask yourself: 'Why me?' It may be an offer you need to refuse.
- *Dependent on the completion of another project*. Even if you are 'guaranteed' that projects you hope to use as data sources will be completed in time for your use, you are strongly advised not to make your own project dependent on them. If slippage occurs, your own research will be held up or even scrapped.
- *Unethical*. Avoid taking on projects that can damage other people physically, emotionally or intellectually. Refuse to take on a project that forces you to breach confidentiality or trust. When using interviews, observation or surveys, you will need to pay particular attention to politically sensitive issues such as power relationships, race, gender and the disclosure of personal information. Ethics are discussed in more detail at the end of this chapter and elsewhere in this book.

Activity 3.4

Consider each of the following 'big' topics and formulate a more focused, narrower research project from each of them:

- Communication in the workplace
- Mergers and acquisitions
- Health and safety
- Teenage pregnancies
- Equal employment legislation

PLANNING THE PROJECT

It may seem obvious that all research projects should be carefully planned, but it is surprising how many researchers rush forward into data collection without a plan of campaign. Disaster is the inevitable result. Planning also helps with time management, one of the greatest problems when work and research commitments compete. There are many ways of planning a project and presenting the plan. One technique is to make use of a table, which sets out the tasks and the planned dates for their completion. A better approach is through the use of a Gantt chart (see Figure 3.3) through which you not only specify tasks but whether they are going to be completed in sequence or in parallel. Project management software such as *Microsoft Project* not only generates various graphics such as Gantt charts, but also allows you to specify timescales for the completion of each task.

LOCATING THE LITERATURE

This is not simply a sequential process of first finding the literature that you need and then reviewing it. As we shall see, it is more usual for this to be an iterative process, with locating, reviewing and then more searching. There are, essentially, two main areas to be searched:

- The literature dealing with the subject of your research.
- The literature on research methodology and data collection methods.

The challenge is that there is so much information available. Table 3.2 offers a brief overview of some of the sources.

According to Hart (2001), the keys to conducting a successful search of the literature are: planning, understanding the ways in which information is organized and made available, maintaining records, and extracting information from useful sources, including the main arguments, theories, concepts and definitions. We will look at some of these issues in more detail. In doing so, various sources, primarily websites, will be mentioned.

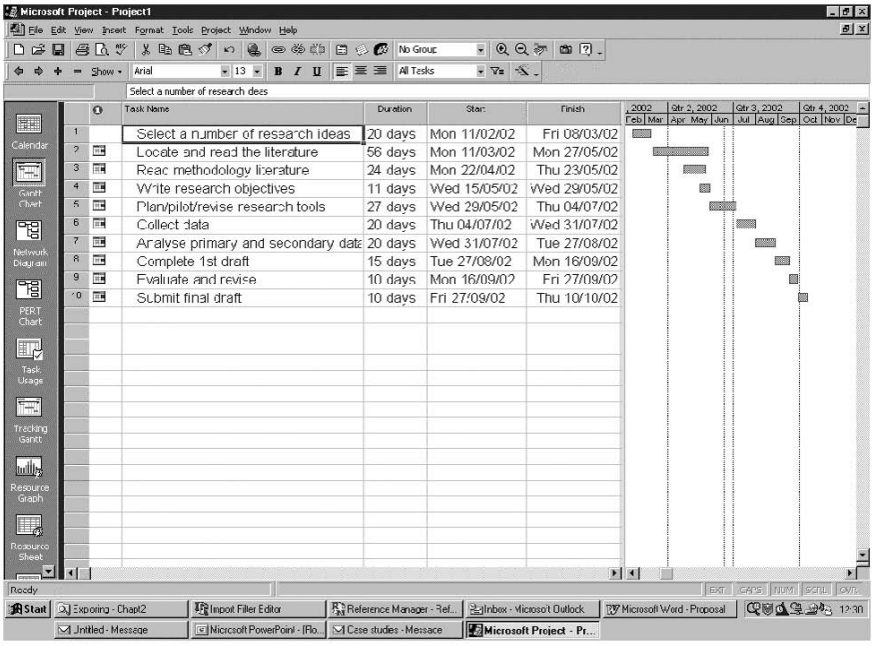


FIGURE 3.3 PLANNING SCHEDULE FOR A RESEARCH PROJECT USING MICROSOFT PROJECT

Search planning

At the outset you may have only a general notion of your research theme (for example, performance management in your company, the effectiveness of mixed ability teaching, public attitudes to the integration of schizophrenic patients into society). This is obviously where you will start. It may happen, however, that your reading takes you down a completely different path. This is acceptable, as long as it results in a coherent topic for your research. Alternatively, your initial theme may appear too wide (for example, brands in marketing), and through reading the relevant literature you may be able to focus on a specific area of the topic which is actually feasible to implement (for example, getting a new brand image accepted and promoted by an organization).

Planning the literature search is not like getting on a train and travelling from A to B. You may have a general intention of getting to B, but your route may involve several modes of transport (sources) and you may find yourself travelling sideways or sometimes backwards! Indeed, you may even decide when on your journey that you intend to travel to C instead! Planning, then, means aiming for a destination (even though this goal may change) and knowing where the modes of transport and timetables are. To modify Hart's (2001) analogy of trawling and mining, the research process is about touring (looking around) and camping (stopping to explore in more depth), as Figure 3.4 shows. Notice that the travelling

TABLE 3.2 AN OVERVIEW OF LITERATURE SOURCES

Sources where research and information is published	Sources and organizations providing publications	Tools for searching published works and data
Textbooks	Academic libraries	Library catalogues
Articles	Public libraries	Subject indexes
Theses	National libraries	Subject abstracts
Government publications	Specialist libraries	Bibliographies
Legal and professional publications	Museums	Encyclopaedias
	Archives	Guides to the literature
Trade literature	Special collections	Internet directories
Conference papers	Political parties	Internet search engines
Monographs	Commercial organizations	
Statistics	Trusts	
	Internet	

Source: Adapted from Hart, 2001

process involves an enticing journey around the literature. But camping involves a more discriminating approach, selecting, synthesizing and analysing in more detail. By the end of the literature search journey, you will be familiar with the history of the subject, the key sources and authors, and methodological approaches, theories and findings. Above all, you will be familiar with the problems, debates, arguments and uncertainties within the territory, and these should begin to clarify your own concerns, objectives and research focus.

Managing information

It is all too easy to be enthusiastic and motivated in searching the literature, but sloppy in storing your findings. Without an accurate, consistent and searchable means of storing your literature search data, your efforts will lack the reward that they deserve. The key is the maintenance of a research log. This can be paper-based or a computer file, depending on which you are most comfortable with. The research log could contain sections on:

- Search parameters – details of your main subject focus and the keywords that describe it.
- Search logs – information on what you have searched, when, how and with what results.
- Contact details of people, organizations, Internet newsgroups, online discussion groups, etc.
- Inter-library loans, including what you ordered and when.
- CD ROM and Internet sources.
- Bibliography.

Maintaining an accurate and searchable bibliography is important for a number of reasons. First, it means that you have a printable bibliographical record at the end of your research project. Secondly, keeping a searchable record allows

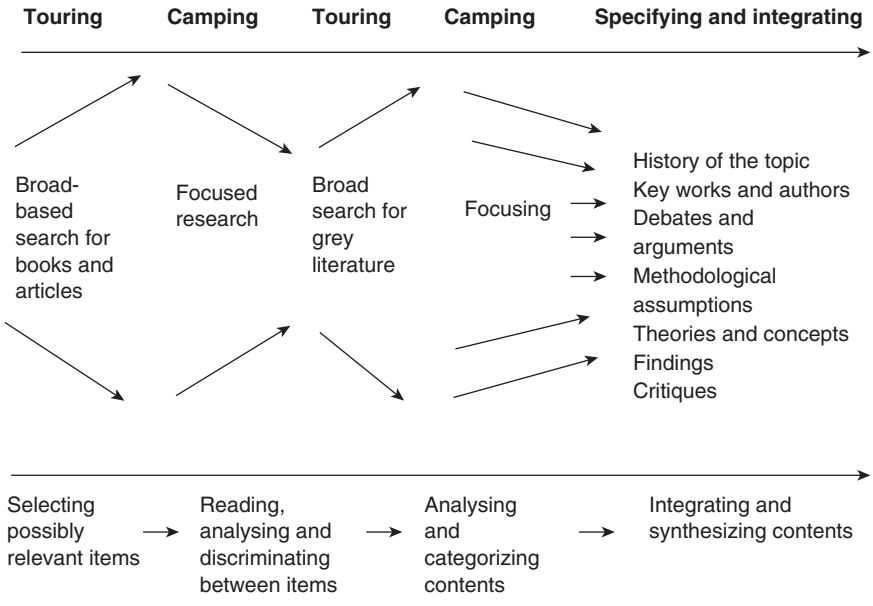


FIGURE 3.4 TOURING AND CAMPING TO ACQUIRE, ANALYSE AND SYNTHESIZE INFORMATION (ADAPTED FROM HART, 2001)

you to locate all your references on specific issues (topics, authors, dates of publication etc.) when you are writing up your research. This is much easier to carry out if you maintain computerized records. There are a number of bibliographic software products on the market that allow you to store records in a database, create bibliographies and even conduct Internet searches of online and Web databases.

Activity 3.5

You can download trial versions of two bibliographic software products from the Adeptscience website at:
http://www.adeptscience.co.uk

Look for Reference Manager and EndNote.

For the specifications of the two applications see:
http://www.lib.ucdavis.edu/healthsci/win-bibmanagers.html

For a tutorial on using Reference Manager see:
http://www.ion.ucl.ac.uk/library/rm.htm

For a tutorial on using EndNote see:
http://www.endnote.com/support/en4tutorial.asp

Locating reference materials

There is no strict order in which you have to locate source materials. Creswell (1994), however, suggests that the following order is helpful:

- *Journal articles.* But ensure that you concentrate on peer-reviewed journals. This is not to say that you should always avoid, say, professional magazines or periodicals. Articles in credible journals, however, are scrutinized for their relevance, originality and validity by other academic researchers. Hence, the quality and reliability of such articles, while not guaranteed, is more assured.
- *Books.* Begin with research monographs that are summaries of the academic literature on a particular subject. Then look at complete books on a single topic or ones that contain chapters written by different authors. Note, however, that textbooks quickly become dated, so it is best to use them in conjunction with other sources.
- *Conference papers.* These often contain the most up-to-date information and developments in research.

In contrast, Hart (2001) advises that the best start is delving into a library's quick reference section, where you can find sources such as guides to the literature, dictionaries, directories and encyclopaedias. After this, the focus becomes one of using books and bibliographies in more detail. Clearly, there are alternative approaches! Try dipping into the reference sources and see what you find the most rewarding.

Books

The main source for locating relevant books is the Online Public Access Catalogue (OPAC) of a library. These can usually be searched by subject, author, title and key words. They also often offer access to other online catalogues, and gateways. Using the Internet, you can access libraries across the world, as the next Activity shows.

Activity 3.6

Access the BUBL Link public access catalogues and locate the online public catalogues in your own country at:
http://link.bubl.ac.uk/libraryopacs/

Another source of useful information is Amazon, which not only sells books but also often provides descriptions of their contents. See: http://www.amazon.co.uk

When browsing in a library or bookshop, a useful evaluation procedure for deciding on whether a book is worth borrowing or purchasing is to look at a number of features, including the:

- Title and subtitle – are they relevant?
- Preface – does it cover your subject area or at least an element of it, and is it at the right level?
- Contents list – does it offer material on your topic?
- Publisher – is the organization respected for publishing quality texts in your field?
- Bibliography – is there one, and do the references look familiar, at the appropriate level and 'on topic'?

Bibliographic services

Bibliographies are gigantic databases of what is in print and what has been printed, and can be used for locating sources on topics or by specific authors. Take a look at two major national bibliographies in the next Activity.

Activity 3.7

Explore and evaluate the bibliographic services of the British Library at:
http://www.bl.uk/services/bibliographic.html

Also take a look at the National Union Catalogue of Manuscript Collections at:
http://leweb.loc.gov/coll/nucmc/nucmc.html

Journal articles

There are, essentially, two types of journal article: academic articles written in peer-reviewed journals, and articles published in professional journals (that are not usually peer-reviewed). The articles you need can be accessed through using either an indexing or abstracting tool. A keyword search of an index will give you a list of articles, authors, title of article, date of publication and journal title. Examining the article title or knowing the reputation of the author may allow you to evaluate whether the article is worth reading. Abstracting services, however, also give you a brief synthesis and summary of what the article is about, which is much more helpful. Since many (but unfortunately not all) journal articles are now published online, it may even be possible to access the article immediately and print it off (sometimes, however, this will depend on whether your institution subscribes to the journal title). Hart (2001) advises that it is best to search the multi-subject indexes and abstracts first, and subject-specific abstracts (if there are any) last. See the next Activity for an example of a multi-subject index and a subject-specific index.

Activity 3.8

For examples of *multi-subject indexes* see:

The Cambridge Scientific Abstracts and look for Electronic Products at: http://www.bhinet.co.uk/
Catchword at http://www.catchword.com/
Ebsco online at http://www-uk.ebsco.com/home/

For a *subject-specific index* see:
ScienceDirect at: http://www.sciencedirect.com/

Many journals are now either being published in both paper-based and electronic format, or just as e-journals (the latter being increasingly the case). There are also a growing number of gateways to these e-journals that provide access to the full-text articles (either free of charge or for a subscription). Explore some of these very useful gateways in the next Activity.

Activity 3.9

Take a look at the BUBL Journal collections at:
http://bubl.ac.uk/journals/

And then at Ingenta journals at:
http://www.ingenta.com

For a directory of electronic journals see:
http://arl.cni.org/scomm/edir/index.html

Grey literature

This is the sort of published and unpublished material that cannot be identified through the usual bibliographic methods. A growing and significant example here are websites, but grey literature also includes academic theses and dissertations, newspaper and magazine articles, editorials, materials produced by business and trade journals, reports, and publications by clubs and societies.

Activity 3.10

For a search engine providing access to technical grey literature see the Magic Project at:
http://www.magic.ac.uk/

Links to Internet sources on grey literature are at:
http://www.library.drexel.edu/netresources/subject/greylit.html

Official publications, statistics and archives

These can be of enormous value to the researcher, but it will depend, of course, on the extent to which a government collects these kinds of data, and the level of access provided to the public. Of the kind of material that is available, an increasing amount is finding its way onto the Web. The next Activity provides links to some multi-subject gateways, sources for statistics and links to data archives.

Activity 3.11

UK statistical sources can be found at:
http://www.statistics.gov.uk/

European Commission statistics can be viewed at:
http://www.europa.eu.int/comm/eurostat/

A source of data archives can be found at:
http://ssda.anu.edu.au/
http://www.psr.keele.ac.uk/data.htm

Using citations and reviews

A citation index records the references made in other works to an author or source. By examining the scale to which an author has been cited, you can quickly see who the acknowledged authorities are in a particular field. Citation indexes also provide reviews of books. Book reviews can also be found in abstracting services, in professional magazines and journals as well as in the academic journals.

Activity 3.12

Point your Web browser to the Institute for Scientific Information (ISI) site at:
http://www.isinet.com/

Find the Journal Citation Report on the Web, which provides access to statistical data to determine the relative importance of journals within their subject categories.

Using authors

Consider making use of networks to access literature sources. It you have been impressed with a particular author's work, why not try contacting her or him? These days, many people are quite easy to locate through their organization's website. You can ask them if they have published anything else in your field of interest, or if they are writing something new at the moment. You may be lucky enough to receive copies of articles, drafts of work in progress, or at least new references. On the other hand, they may be too busy to reply to you, so do not be disappointed!

Joining professional associations

You can join relevant professional associations, many of which publish their own professional journal, and hold conferences or seminars. Some even have their own libraries that can provide a rich source of material in the field.

Using your organization

Do not forget that you can also make use of your own organization as an important source of data. This will include internal documents. But note that the academic literature should underpin the theoretical elements of the project. Using institutional or company documents is valid for providing background information and supporting detail, but they should not carry the main burden of the theoretical argument or analysis.

REVIEWING THE LITERATURE

The critical review of the literature provides the foundations of your research. Not only does it inform and refine your research objectives (for example, are they topical, worthy of research, original?), it provides a benchmark against which you can compare and contrast your results. One of the features of any project is that it should enable you to demonstrate a critical awareness of the relevant knowledge in the field. A comprehensive review of the literature is essential because it:

- Provides an up-to-date understanding of the subject and its significance and structure.
- Identifies the kinds of research methods that have been used.
- Is informed by the views and research of experts in the field.
- Assists in the formulation of research topics, questions and direction.
- Provides a basis on which the subsequent research findings can be compared.

The literature review is not something you complete early in the project and then drop. It is likely to continue almost to the writing up stage, especially since your own research may generate new issues and ideas that you will want to relate to the literature. As we saw in Chapter 2, if you are adopting a deductive approach to your research, your literature review will help to provide a source for the focus of your research, including aims, objectives and hypotheses. Conversely, if you favour a more inductive approach, you may begin with the collection and analysis of data, a process that leads to questions that are then addressed through engagement with the literature.

The critical review

A review can involve a narrative or description of an article or other piece of work. A *critical* review, however, is much more than this and it is important to

TABLE 3.3 SKILLS FOR CRITICAL ENGAGEMENT WITH THE LITERATURE

Skill	Actions	Description
Analysis	Select, differentiate, break up	Dissecting data into their constituent parts in order to determine the relationship between them
Synthesis	Integrate, combine, formulate, reorganize	Rearranging the elements derived from analysis to identify relationships
Comprehension	Understand, distinguish, explain	Interpreting and distinguishing between different types of data, theory and argument to describe the substance of an idea
Knowledge	Define, classify, describe, name	Describing the principles, uses and function of rules, methods and events

Source: Adapted from Hart, 1998

understand the difference. Hart (1998) argues that a reading of the literature should pose questions such as:

- What is the purpose of the study?
- What is the focus of the study?
- What types of data were collected?
- How were the data managed?
- What analytical approach is used?
- How is validity addressed?
- How are ethical issues handled?

Operationalizing the issues listed above means using sets of tools to analyse and evaluate the literature. This means developing a complex set of skills, acquired through practice. Table 3.3 provides a brief overview of the types of skill involved.

In terms of structure, Creswell (1994) suggests that a literature review should comprise five components:

- An Introduction, informing the reader about how the review is to be organized and structured.
- Review of Topic 1, addressing the literature on the independent variable or variables (the influences on the dependent variable or subject, upon which the research is focused). Note: we will explore descriptions of dependent and independent variables in more detail in Chapter 4.
- Review of Topic 2, the literature on the dependent variable. If there are multiple dependent variables, devote a sub-section to each one, or focus on a single important dependent variable.
- Review of Topic 3, the literature that relates the independent variable to the dependent variable. Creswell warns that this section should be relatively short and should focus on studies that are extremely close in topic to the proposed study. If nothing specific has been written on the topic, then review studies that address it at a general level.

- Provide a summary of the review, highlighting the most significant studies and the key themes that have emerged.

This last point is vitally important. It is not enough to simply read around a subject or theme. You must produce a synthesis of subjects or related subjects in the form of an increasingly *focused argument* or set of concerns (recall Figure 3.4). It is the difference between describing a menu – 'first we tried the appetizers, then we ate the main course, after which we sampled the desserts' – and producing the sort of *critical* evaluation of a restaurant's fare that you would read in a gastronomic magazine. Within the five-step structure suggested above, any critical review should also incorporate:

- An assessment of the strengths and weaknesses of some of these theories.
- A clear understanding of the topic.
- A citing of all key studies in the subject territory.
- A clear indication of how the review links to your research questions.
- A definition of the boundaries of your research.
- A selection and synthesis of existing arguments to form a new perspective.
- Through gradual refinement, a clear demarcation of the research problem.

The key word here is 'refinement'. Recalling Figure 3.4, touring the literature, but also pausing to focus on areas that have emerged as important, means that the discussion is gradually refined down to a set of core issues and arguments. These, then, provide the basis for the formulation of research questions and the focus of the research.

Hart (1998) warns that, in evaluating a piece of research, the researcher must be aware of the methodological tradition from which it emanates – even if the researcher is not sympathetic to that tradition. For example, it is not enough to criticize a quasi-experimental research study for taking a quantitative approach to data collection, since this is what one would expect. A more valid argument would be that the research design was not matched to the research objectives, or that assertions made for the study were insufficiently supported by the data or analysis.

Positioning the literature review

Should your review of the literature come at the beginning of your report or dissertation, in the middle or at the end? Creswell (1994) offers three possible locations: in the *Introduction*, as a *separate section* and as a *final section* in the study. Table 3.4 provides a brief summary. For some qualitative studies, for example, the literature can be discussed in the *Introduction* to 'frame' the subject being studied. Using a separate literature review section towards the beginning of a study is typical of a quantitative approach. The purpose of this section becomes to provide a basis on which questions and hypotheses can be based, and can influence the design and direction of the research. In contrast, in some types of

TABLE 3.4 RESEARCH METHOD AND USES OF THE LITERATURE

Research method	Use of the literature	Comments
Qualitative studies: all types	Used in Introduction to 'frame' the problem	Some literature must be available
Quantitative	Located as separate 'review of the literature' section at beginning of study	Helps to generate research questions and hypotheses; also used at end of study against which results compared
Qualitative: grounded theory	To compare and contrast theories generated from the data with theories in the literature	The literature does not guide or direct the study, but becomes an aid once patterns emerge from the data

Source: Adapted from Creswell, 1994

qualitative research, such as the use of grounded theory, theoretical perspectives are developed inductively from the data itself. Any literature review is created towards the end of the research, providing a benchmark against which the results can be compared.

WRITING THE PROJECT PROPOSAL

There are, essentially, two types of proposal:

- An **organizational** proposal, written to gain funding for a project or at least to elicit support and commitment from a project sponsor.
- An **academic** proposal, a plan for conducting research as part of an academic programme of study.

Organizational proposals

There are two main types of organizational proposal, comprising those that are written:

- In response to a request for proposals or 'invitations to bid' from, say, government agencies or companies.
- For submission to an internal organization or department, often in response to a request for help with a problem or a need to improve a product or service.

Whichever kind is written, White suggests that a proposal is a 'sophisticated advertisement of approaches and abilities' (1997: 218) and not just a description of how a project could be tackled.

To bid or not to bid?

Many proposals are written within tight timescales and in the knowledge that other individuals, teams or organizations are competing for the project. In deciding whether to respond to any request for proposals (RFPs), you will need to take a view of:

- Whether you possess the necessary expertise to respond.
- The number of bids you are competing against and the likely strength of opposing bids.
- The number of bids that will be funded.
- Whether all bids are to be fairly evaluated, or whether the process is possibly biased.

These criteria will help you to undertake a risk assessment, weighing up the probability of success and the potential size of the funding if successful, against the time it will take to write the bid.

Preparing to write the proposal

Figure 3.5 illustrates a series of steps that should be observed in writing any project proposal. We will look in detail at each stage.

Review the RFP specifications Take another, careful, look at the request for proposals and make sure that they are complete, consistent and clear. Some RFPs may also contain information on how responses to each of the specifications will be scored – for example, some requirements may be more important to the project than others and receive a higher weighting. If anything in the RFP is unclear, then it is legitimate to contact the creators of the RFP (preferably in writing) and ask for clarification.

Develop a plan of action This should include a rationale for the project, the key research objectives and a plan of how these are to be met, that is to say, the research methodology. Take care to show that the project has value. To do this apply the invaluable 'so what?' test. This means looking at your proposed project outcomes and asking yourself to evaluate honestly whether they have any real worth in terms of practical application or contribution to knowledge.

Determine the human resource requirements of the project Those who commissioned the proposal will be keen to evaluate not only the number of people committed to the project, but also their experience and qualifications. If any element of the project is going to be outsourced to a third party, then again, evidence should be provided of their 'fitness for purpose'.

Develop the project graphics This will be particularly important for more technical projects, but should always be considered for research projects of any kind. Graphics, particularly flow diagrams showing how different elements of a project relate to one another, can be easy to read and understand. Ensure that if

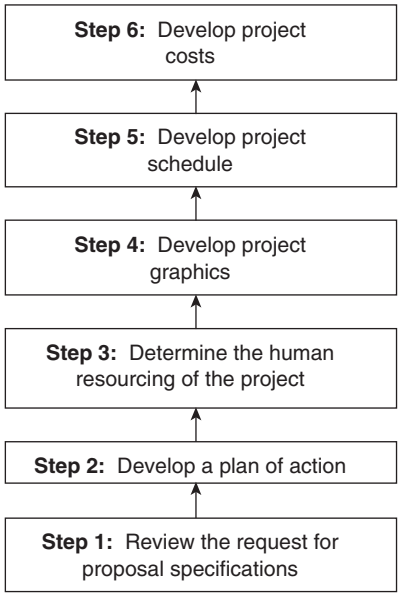


FIGURE 3. 5 PROPOSAL DEVELOPMENT FLOWCHART (ADAPTED FROM WHITE, 1997)

graphics are used, there is a clear description provided in the text (such as the way in which this text is describing Figure 3.5).

Develop a project schedule A project schedule such as a Gantt chart (recall Figure 3.3) provides a list of the main project tasks and the dates for their completion. These tasks could include arrangements for seeking permissions from an ethics committee, preliminary arrangements for getting research subjects, a timetable for constructing research tools, analysing data and writing the research report.

Develop project costs White (1997) warns that this is more an art than a science because there are so many variables to take into account. Particular care needs to be taken when estimating labour costs since these often eat up the bulk of a research project's budget. Since the research process is rarely linear and smooth, it is prudent to add in some contingency costs in case unexpected results emerge that require further research. The less experienced the research team, or riskier the project, the higher the contingency costs should be. Given that commissioning organizations are highly unlikely to pay for contingency costs, these will have to be built into (i.e. hidden!) in general costs. Of course, the higher your overall costs, the less competitive your bid.

Preparing the final document

In writing the final project proposal, try to ensure that the document is internally consistent, and that there are no gaps or contradictions between objectives and

research processes. Remember that once the proposal is submitted you are committed to it, so you need to ensure that all errors have been eliminated. It helps, then, to elicit the assistance of some experienced reviewers who should be asked to answer the following questions:

- Is the proposal clear and understandable?
- Is the proposal realistic?
- Is the proposal consistent with the bid specifications?

Also, get an appropriate person to review the costings. White (1997) makes the useful suggestion that you should do a hand calculation of all columns and rows of financial data, even if they were calculated by computer (in case a decimal point has slipped!). When you have finished the proposal, print it off and read it one more time.

Success or failure?

It is worth having a strategy in place for whether the bid is approved or rejected. If the latter, Williams (1996) offers the advice that you should try to get feedback on the proposal from those who have evaluated it. This may allow you to improve the bid if there are later opportunities for resubmitting it. If the bid is successful, you may be offered less funding than you asked for. At this point you need to decide whether to accept this funding, or, if possible, renegotiate the outputs of the project at a lower level.

Academic proposals

The structure of these proposals is normally specified by the academic institution. An example is given in Table 3.5, where you will note the weight given to academic theory and underpinning.

THE ETHICS OF RESEARCH

We will deal with a wide range of ethical issues that are particular to specific research methods in later chapters. Here, we will examine some of the ethical considerations of more general significance. Ethics is a philosophical term derived from the Greek word *ethos*, which means character or custom. The ethics of research concern the appropriateness of the researcher's behaviour in relation to the subjects of the research or those who are affected by it. Since research in the real world inevitably deals with people and the things that affect them, ethical issues can arise at the planning, implementation and reporting stages of research. Some ethical issues are obvious and a matter of courtesy and common sense but others are less clear. As Korac-Kakabadse et al. (2002) suggest, one of the challenges of ethical behaviour is that it lies in a 'grey zone' where clear-cut dichotomies between what is right and wrong may not exist.

TABLE 3.5 TYPICAL STRUCTURE FOR AN ACADEMIC PROPOSAL

Section	Contents
Working title	Describes the breadth and depth of the topic and gives an indication of the methodology to be used (e.g. case study, evaluation, etc.)
Introduction (abstract)	A summary of the research topic, describing the core problems or issues, the gaps in the research and how the research will address them
Aims	General statements on intent and direction of the research
Objectives	Clear and measurable statements of intended outcomes
Justification	Rationale for the research with reference to gaps in current knowledge, and potential application of results
Review of the literature	Describes the history of the topic and key literature sources; illustrates major issues and refines focus to indicate research questions (qualitative research) or hypothesis (quantitative research)
Methodology	Justifies methodological approach, including data collection and analytical techniques; use of quantitative or qualitative methods; choice of research approach and paradigm; anticipation of ethical issues; how the data will be analysed
Work schedule	A timetable for completing the research, indicating tasks and timescales
References	Bibliography of works cited in the proposal
Related material	For example, letters of support for the research, agreement to collaborate from interested institutions

Source: Adapted from Hart, 1998

Ethical issues arise even at the initial access stage, where the ambitious researcher can unwittingly or otherwise put pressure on people to become participants. If you are a practitioner researcher conducting a project within your own professional context it may be especially tempting to cajole colleagues who may feel obliged to cooperate. The key to ethical involvement is not just obtaining consent but *informed* consent. This includes explaining to gatekeepers such as project sponsors and the participants:

- The aims of the research.
- Who will be undertaking it.
- Who is being asked to participate.
- What kind of information is being sought.
- How much of the participant's time is required.
- That participation is voluntary.
- That responding to all questions is voluntary.
- Who will have access to the data once it is collected.
- How anonymity of respondents will be preserved.

Informed consent may be given through a verbal agreement, but a written statement is probably best. It is also important to keep participants informed if the aims of the research change during the project.

Professional researchers in the fields of sociology and psychology often have a set of ethical guidelines to adhere to, as do members of some professional associations or those working in professional contexts. In the UK health service, for example, many research ethics committees have been established, their work coordinated by the Central Office for Research Ethics Committees. Activity 3.13 provides some useful websites that illustrate different sets of ethical guidelines.

Activity 3.13

Compare and contrast the ethical principles stated at the following sites, noting any difference between codes of conduct for researchers and consultants.

American Psychological Association at:
http://www.apa.org/ethics/code.html

British Educational Research Association at:
http://www.bera.ac.uk/guidelines.html

Central Office for Research Ethics Committees at:
http://www.corec.org.uk/

Institute of Management Consultants (see Ethics and Discipline, under 'Our Standards') at:
http://www.imc.co.uk/index3.html

Table 3.6 summarizes some of the ethical issues the researcher must be aware of and plan for. The following Case Study provides you with an opportunity to apply some of these principles, and illustrates some of the complexities of trying to adopt an ethical stance in 'real world' situations.

Case Study 3.1 Ethical dilemmas in research

A financial services company implements a new information technology system that gives e-mail capability to all employees, irrespective of grade. After 12 months, company directors are anxious to know if the investment is providing a financial payback. One indicator is whether the e-mail facility is improving inter-employee communication and general productivity. Hence, they want an analysis of e-mail traffic. An external researcher is commissioned to conduct the study.

She decides to use a blend of quantitative and qualitative methods. On the quantitative side, negotiating access to the information poses no problems, since all she has to do is contact the head of information services. Given that this is a legitimate

(Continued)

TABLE 3.6 A CHECKLIST OF ETHICAL ISSUES

Ethical issue	Description
Privacy	The right not to participate. The right to be contacted at reasonable times and to withdraw at any time
Promises and reciprocity	What do participants gain from cooperating with the research? If promises are made (such as a copy of the final report) keep them
Risk assessment	In what ways will the research put people under psychological stress, legal liabilities, ostracism by peers or others. Will there be political repercussions? How will you plan to deal with these risks?
Confidentiality	What constitutes the kinds of reasonable promises of confidentiality that can be honoured in practice? Do not make promises that cannot be kept
Informed consent	What kind of formal consent is necessary and how will it be obtained?
Data access and ownership	Who will have access to the data and who owns it? Make sure that this is specified in any research contract
Researcher mental health	How will the researcher be affected by conducting the research? What will they see or hear that may require debriefing or counselling?
Advice	Who will the researcher use as a confidant(e) or counsellor on issues of ethics during the research?

Source: Adapted from Patton, 1990

company project, commissioned by the executive board, he is obliged to give the researcher free access to whatever information she requests.

For the qualitative side of the research she wants to interview a sample of 40 employees. Believing that imposing herself on these people would be unethical, she writes to all of them individually requesting access, and provides details of the purpose of the research, how the information is being collected, and who will read the final report. She had made a request to the executive board that she should be allowed to provide a summary of the final report to all respondents but this was refused. Despite her reassurances, only 12 of the original sample agree to being interviewed, most excusing themselves on the basis that they are too busy. One option would be to obtain an instruction from the managing director, ordering everyone to cooperate. She decides, however, that, not only would this be counterproductive, it would be unethical on a number of grounds not least because the responses would no longer be voluntary. Eventually, she decides that these 12 in-depth interviews will be sufficient to yield high quality data.

Having set up the interviews, the researcher first turns to the quantitative analysis of the e-mails, which she intends to place into a number of categories. However, as the research progresses, she discovers a significant number of personal e-mails, including jokes, a betting syndicate, plans for illicit liaisons and inflammatory comments about senior managers and the executive board!

The researcher now faces a difficult ethical dilemma. She decides to include general descriptions of the personal e-mails in her report but not to reveal the names of individuals (although it will not be difficult for the company to trace them given that it now has an e-mail audit trail). She also decides that she will ask

(Continued)

some questions about personal e-mails in her interviews to gain an employee perspective. Before doing this, she takes another look at her letter to the interviewees and the description of her research. She decides that the description, 'To investigate the purpose of e-mail traffic' is still valid and an additional letter flagging the new 'personal e-mail' probe is not necessary. Participants will still be given the assurance that they can refuse to answer any question and that their responses will be anonymous.

Activity 3.14

Examine Case Study 3.1 and consider the following questions:

1 Is the research conducted by a researcher or a consultant? Does it make any difference to the ethical issues involved?
2 Have sufficient steps been taken to safeguard ethical principles? Are there any additional steps that you would take?
3 Should the research be abandoned before the qualitative stage?

Suggested answers are provided at the end of the chapter.

SUMMARY

- A good research topic must have the potential for demonstrating theoretical credibility, allow access to the relevant data, provide a symmetry of potential outcomes, and be congruent with your own interests, capabilities and career aspirations.
- To generate ideas for a research topic you could look at examples of projects completed by others, or ideas could emerge from your reading of the literature or by a brainstorming process.
- Before starting the project, produce a plan to show when and how you intend to conduct your research including data collection, analysis and the writing up process.
- An important step in any research project is the literature review. This may assist in the formulation of research questions or topics (deductive approach) or illustrate and illuminate research findings (inductive approach).
- Before starting many projects you may be required to write a project proposal. Make sure that, if there is a request for proposals, your bid matches the specifications accurately. Get your proposal evaluated by others before submission.
- Ensure that ethical principles are catered for in the research, including the privacy of respondents and their anonymity (if this has been guaranteed).

(Continued)

Summary of web links

http://www.adeptscience.co.uk
http://www.apa.org/ethics/code.html
http://arl.cni.org/scomm/edir/index.html
http://www.bera.ac.uk/guidelines.html
http://www.bhinet.co.uk/
http://www.bl.uk/services/bibliographic.html
http://bubl.ac.uk/journals/
http://link.bubl.ac.uk/libraryopacs/
http://www.catchword.com/
http://www.corec.org.uk/
http://www.endnote.com/support/en4tutorial.asp
http://www.europa.eu.int/comm/eurostat/
http://www.imc.co.uk/index3.html
http://www.ingenta.com
http://www.ion.ucl.ac.uk/library/rm.htm
http://www.isinet.com/
http://lcweb.loc.gov/coll/nucmc/nucmc.html
http://www.library.drexel.edu/netresources/subject/greylit.html
http://www.lib.ucdavis.edu/healthsci/win-bibmanagers.html
http://www.magic.ac.uk/
http://www.psr.keele.ac.uk/data.htm
http://www.sciencedirect.com/
http://ssda.anu.edu.au/
http://www.statistics.gov.uk/
http://www-uk.ebsco.com/home/

Further reading

Jankowicz, A.D. (1991) *Business Research Projects for Students*. London: Chapman & Hall. Very readable and contains a practical guide to selecting and planning research projects.

Hart, C. (2001) *Doing a Literature Search*. London: Sage. An essential guide that includes plenty of practical advice and also a host of useful online resources.

Hart, C. (1998) *Doing a Literature Review*. London: Sage. Another essential source that justifies the importance of the literature review and demonstrates the review process.

Suggested answers for Activity 3.14

1 Since the researcher is hired from outside the organization, it is probably fair to call her a consultant. Does this make a difference to the ethical stance adopted? Well, possibly, yes. Researchers, for example, may be measured against the code of conduct of their relevant professional association. Consultants may also have a professional association, but also have to answer to the research sponsor or manager who will tend to be more concerned with 'results'.

2 The consultant has adopted a number of ethical safeguards, including asking for participation and providing information about the purposes of the research.

3 If this project was being conducted in a purely research context it would probably have to be abandoned. But for the consultant, while needing to give due weight to stakeholders' interests, the interests of the client come first.

PART B

Research Methodology